Overview

Project: MkDocs Quiz Plugin

Epic: Requirements

Deliverable: Interview Transcript

Version: 1.4.2

Prepared by: Incze Antonia, Maniu Diana

Interview Preparation

Invitation email draft

Dear Prof. Dr. Stefan Zander

My name is Incze Antonia. Me and my team are in the process of developing an MKDocs plugin that aims to convert Markdown files into interactive quizzes. Its purpose is to make learning materials more engaging for both students and educators.

As part of my user research, I would like to invite you to a short interview to better understand how professors like yourself would use such a tool and what features would be most useful from an educator's perspective. Your insights would be invaluable in our project's development.

If available, I would be happy to schedule the interview either in person or online this Friday, October 31st, after 12 pm, or we can discuss availability in the coming week, and I will make arrangements accordingly.

Thank you for your consideration.

Kind regards,

Incze Antonia

Intro: briefly describe project
Prepared Questions

1. How do you currently assess or engage students? (eg. quizzes, polls, other interactive methods)
2. Have you used any digital quiz or learning tools?
   Follow up if yes: What do you like about them? What do you dislike about them?
3. Are you interested in integrating quizzes directly into your online materials? What key features would you expect or need from said quizzes?
4. How do you imagine students would interact with these materials?
5. What type of data would you want to see regarding student performance on the quizzes?
6. How important is visual customization and accessibility for the plugin?
7. Overall, what would make a quiz plugin like the one described genuinely helpful to you as an educator?

Transcript

Host: Antonia Incze, assisted by Diana Maniu
Participant: Prof. Dr. Stefan Zander
Date: 31/10/2025
Workspace: BigBlueButton
Recording Link: https://recordings.h-da.de/r/5290fdd5762667adf7f4f4255d7ee710ad50da25-1761908651590/index.html

[Stefan Zander]: The session is recorded right now.

[Antonia Incze]: Ok, so, I will start with the first question: how do you currently asses or engage your students?

[Stefan Zander]: In general or in a specific setting?

[Antonia Incze]: Mainly outside of courses.

[Stefan Zander]: Outside of courses... and what do you mean with "How do I engage", do you mean in terms of asking whether there are open issues or informing students about what is required from them?

[Antonia Incze]: I am mainly talking about the specific... I know the word in German, not in English right now - "Inhalt" of the course, like if you send quizzes or polls or any other interactive methods to engage your students.

[Stefan Zander]: Okay, yes. So, usually what I'm going to do is, when there's the registration phase at the beginning of the semester and I have a list available of the persons who would like to take my course or who had registered for my course, I will prepare an email, giving them the most important information, like when does the first lecture start, when is the kick-off, when will the practical course start, and I also include a link to a webpage with information about my courses. So, on this webpage there is a course listed, together with information about how the course will be conducted during the semester, about the module, about the milestones, about the contents like the topics we are going to talk and learn about, and also about the practical sessions.

[Antonia Incze]: And during the semesters, do you send them any kind of quizzes, for example on Moodle or Google Forms.

[Stefan Zander]: No.

[Antonia Incze]: No?

[Stefan Zander]: Not at all. To be honest, I hate Moodle quizzes. It is so cumbersome to use them and to offer quizzes so, yeah, I hate it to be honest, to be really honest.

[Antonia Incze]: What exactly makes it so cumbersome?

[Stefan Zander]: The UI is very bad, the usability is very bad and the third aspect is, if I would; my objective is to use Moodle as little as possible, and if I would create quizzes in Moodle, this would require to integrate it more tightly into my lecture material, and this would refuse the possibility for me to have some kind of flexibility, because I'm tied to Moodle and what they offer. For me, it would be

better to be able to work in my own environment, to have my own tools available in which I author all the course materials and that I could also use to author the quizzes. It would also help me – this is some of my wishes – to integrate the quizzes in my course material, otherwise I could only put a link in my content, in my lecture material, that goes to the Moodle course, and it would require all students to be part, to be registered in my Moodle course. This is something I… this would create a higher workload and with reduced flexibility.

[Antonia Incze]: I see, that was a really good answer, it actually perfectly ties into the next question I had: if you could integrate quizzes directly into your course materials, what key features would you expect from that?

[Stefan Zander]: If I could create quizzes into my course material?

[Antonia Incze]: Yes

[Stefan Zander]: Yes, ok. What features… It's not a feature, it's more like a wish: the best wish for me or the best outcome would be when the students say after they conducted a quiz, that it really helped them, that it created an additional value. So, this is for me the main evaluation criteria, and I hope to achieve that with the following features like: the possibility to have different types of questions, the possibility, second, to closely tie them to the material the questions refer to. I do not want to have a quiz on any other page or on any other system, but I want the students to have a page they- this is the course material, and this is the possibility for you to take the self-assessment, right below the main course material, or the main lecture or session material. The third feature I find really helpful would be to not have kind of a media break or system break, so for instance, if you want to take a self-assessment, you need to call a different system, so do it right where you are. From there, it would be good for students to have some kind of feedback, like how well they performed in those questions, like which questions did you answer correctly, which questions were wrong, so what is your overall performance? Maybe also in relation to other students, or in relation to a baseline, so say, ok, you did the quiz the first time and you were below or in the top 20% for instance, so that the students not only get a feedback regarding what questions did they correctly answer, but also how well was their performance in relation to a baseline, or in relation to the overall students' performance. What would also be great is to have kind of an overview for me and for the students: what self-assessments were taken by a specific student. So, for instance, it says the course has 7 topics and for each topic there is a set of self-assessments, possibilities of modules and we see: ok, there is a student who participated in this and that, so perhaps, what the students will see, what self-assessment modules did I conduct, and for me, who conducted the assessments.

[Antonia Incze]: I see, and regarding further feedback on your side, because what you said so far is a lot of feedback for the students, but I'm also interested in hearing what kind of feedback you would want to see on your side, for example, if there is a specific question that is often got wrong.

[Stefan Zander]: Yes, yes, this is a good idea. This would be a great feature for me to say: ok, what is the… how successful was a specific type of question answered by the students? Was there, for instance, an assessment that had been taken by almost none of the students? So then I can dig deeper and find out what the reasons might be, and on the other hand it would be cool to see, okay, yes, what was the overall performance per question type?

[Antonia Incze]: Per question type or per question?

[Stefan Zander]: Per Question.

[Antonia Incze]: And… From that idea…

[Stefan Zander]: Sorry for the interruption, I am using the active learning theory, so my model is based on the active learning theory and this requires some kind of self-preparation by the students, and under ideal circumstances the self-preparation… should have, or should be conducted before taking the class, before going into class and diving deeper into a specific topic. But I know how I myself, as a student, I had behaved the same way, only a minority of the students took the preparation or conducted the preparation, so it's below 10%. So, for me, it would really be helpful: before I do the actual in class lecture, to have a list and kind of a statistic about what preparation had been done, what assessments had been done by the students before attending the lecture.

[Antonia Incze]: I see.

[Stefan Zander]: Was it clear?

[Antonia Incze]: Yes, I understood. I also wanted to ask how important would it be for you to have customization in the appearance of these quizzes, for example: colors, fonts; so that it can integrate well with the rest of your course.

[Stefan Zander]: At this moment, at this stage of my thinking, I would not need to have a lot of customization possibilities. So, when the rendering of the quizzes complies with the design of the page in which it is embedded, then it would be totally fine.

[Antonia Incze]: I see, so no need for customization.

[Stefan Zander]: No, I do not need to have a specific font or a specific font size or a specific color, for me it would be nice to have a default design, that repeats, so the self-assessment should have the same visual appearance on every page on which it is used, so that you can recognize it and that it seems familiar very soon.

[Antonia Incze]: I see. I want to go back a bit to Moodle. I know you said you really hated it, but is there anything that you do like about it?

[Stefan Zander]: Okay. This is a subjective question. What do I like about Moodle, I mean first of all I like about it that we have it as a system in our university, so to have the possibility or the feature to have a central place in which you can host your learning materials.

[Antonia Incze]: So just that maybe it is integrated in your institution right now, but it doesn't really have anything to do with Moodle as a platform itself, it just has to do with the fact that it's convenient because everyone else in your institution uses it.

[Stefan Zander]: Yes, or is able to use it, but from my experience over the last years, my goal is to use Moodle as little as possible. My objective actually is to go away from it as far as I can and as far as is convenient for the students. Because… the reasons for that is, it limits my thinking and it limits the way… my spectrum of capabilities on how to offer course material and present course material, so the boundaries are too small and this limits my thinking, so when I'm working with my tools like VSCode, when I'm able to offer the things in Markdown, I do not need to invest a lot of thinking in how to offer it and in how to meet this specific requirement, present it in a way that complies with the Moodle requirements and restrictions. In Markdown I can start thinking about the content, and not how to put

it and form it into a technical structure. This is basically the main feature why I want to get away from Moodle and why I want to do my own things, or the things I think better suit my way of working.

[Antonia Incze]: I see. What about accessibility, so what would you mainly use to write these courses and quizzes?

[Stefan Zander]: Do you mean accessibility from my side or accessibility in terms of how the students can conduct the quizzes?

[Antonia Incze]: Both.

[Stefan Zander]: Okay. Accessibility... I am totally happy with when I'm able to offer the quizzes like I would write a program or write a software. I have no specific requirements in terms of accessibility. Uhm... in terms of the students, I'm not sure about that, and I'm not sure about- I need to learn about what students require or what they wish to have. Maybe they wish to have kind of the possibility to increase the font size or to change the contrast, or... to have the possibility to have the text spoken to them, or to have the possibility to maybe have an auto translation, like you have the default language, for instance, it German, but you as a foreign student you want to take the self-assessment but in your own language, like in Romanian for instance, so it would be cool to have an auto translation feature included.

[Antonia Incze]: So, we are nearing the end of my questions, I just wanted to have an overall at the end, so, overall what would make a quiz plugin like this genuinely helpful to you? What would make you really want to use this quiz? This quiz plugin, I mean.

[Stefan Zander]: If it unobtrusively integrates with my current authoring environment, so meaning I could use my Markdown editor not only to write my course material but also to write and author the quizzes. It would further help me to have a clear, concise and simple to use syntax, with only a few extensions and I'm willing to learn those extensions when I see their additional value that is gained from them. It would also be nice to have the possibility to not only record multiple choice questions but maybe also some other types of questions, like fill in the gap, like sorting something, like... it would also be cool to have kind of an attachment, in case based learning, like presenting the student with a certain situation and then the student should write in a few sentences how they would handle the situation. I know there is no right or wrong, but to have a feature in the background like LLM based or pre-trained model or whatever, that then gives them feedback about what might work good and what things they should also consider, or they'd left out of their consideration. I know this is future, future, future work, but I see a lot of additional value in that, but this is also something I could do in person in the lecture, so... yes.

[Antonia Incze]: It's like a nice to have but not a need to have.

[Stefan Zander]: Yes, yes, of course. Not a need to have, it would be a cool feature, but... so, different question types and the possibility to store the quiz source code together with my lecture material, so that I have a single point of truth, in which I can make edits to the material and also to the quizzes.

[Antonia Incze]: Okay. Thank you very much for your insights. That was it from my side, if you have any further questions?

[Stefan Zander]: Mhm, uhm… no, not at the moment, I'm just thinking about whether there is additional information I should give you, but the main points I think we talked about, in terms of the customization, what would really be good is to have some kind of proposals. So, for instance, like providing the possibility to have it like a book, only one question at a time, but there might also be situations in which it's good to have a list, if you only have three or four or five and you would like the student to have a complete overview of all the things. Maybe that could be a nice customization, the language feature would be cool, it would also be cool for the quiz to adapt to the predefined rendering, like light or dark mode, so when you look at MKDocs, you could – in the top, you could switch to dark mode and the quizzes should adapt accordingly. They should also take the default font and the default font size, and it would also be nice to integrate a picture and have a question type, maybe, what do you see there, questions regarding what's depicted on the picture.

[Antonia Incze]: Visual support.

[Stefan Zander]: Pardon?

[Antonia Incze]: Visual support, right?

[Stefan Zander]: Yes, yes, for instance, I'm just thinking about maybe you have kind of a process scheme and there's one part missing and you ask, ok what's missing there? Or you have a process in which you ask, ok, what phase… in what phase would you conduct a user test? And the answer could be in form of a multiple choice or the answer could be in form of writing, but there you have this introductory text, but you could also integrate a picture there.

[Antonia Incze]: Do you also see value in maybe support for mathematical equations? For example, if you want to write, x to the power of 2, you could write x^2 and the program would have it automatically look like in a Latex program.

[Stefan Zander]: For my courses, actually no at the very moment, but I am sure there are a lot of colleagues and a lot of lecturers that would benefit from such a feature, yes. I think it would be cool to have this, yes, this feature. Maybe not at the base system, but to consider it in term of further iterations.

[Antonia Incze]: Yes, thank you very much. If there is anything else…

[Stefan Zander]: Let me think… I don't know if there is a need for immediate feedback. Like… you're doing a multiple choice, and you tick it off and then it says: oh, this is wrong, but then there is the question, ok, where is the assessment aspect?

[Antonia Incze]: We would have it so that the score is at the end.

[Stefan Zander]: Yes, yes, maybe it would be nice at the end… maybe a feature would be to either skip a question, but to also have the possibility to leave feedback on each question, so for instance there was a question that had been skipped quite often by the students, so the students have the possibility to leave feedback. Maybe at question level, but definitely at the end, and maybe there could be a self-evaluation about the quiz so, in terms of: after you have conducted the quiz, please answer these questions, like how helpful have you found the quiz, how good were the questions and so on and so forth, so students have the possibility to provide feedback about the questions and the self-assessment.

[Antonia Incze]: Yes, that sounds like a really good idea. Okay, thank you, these have really helped me and I'm sure they're going to help my colleagues as well to understand what we would need to integrate into our plugin for... like an educator's perspective, and we hope to create the plugin that you need and that your colleagues would also really enjoy.

[Stefan Zander]: Yes.

[Antonia Incze]: Thank you very much for your time.

[Stefan Zander]: Okay, you're welcome. Thank you for conducting the interview.

Notes

state the structure + the time | ask if there are questions

① How do you assess and engage your students?
(mainly outside of courses) quizzes / polls

Registration phase list with participants that want to participate
↳ mail to them with info about the course
↳ not really link: topics, practical sessions.
polls or other stuff

NO MOODLE QUIZZES – he hates them – bad UI/UX

WANT: own environment
integration of the quizzes in his courses

② Key Features

WISH: student's gain additional value from the quizzes

- various types of questions, ~~designed~~ adapted to the content
- no breaking in the system (continuity)
- feedback for students (stats about their performance of the quizzes)
- perf. in relations with other students (no # 12 of 30)

students

- for him to see who conducted what module (eg. 17 modules and student X took the first 7 of them)
- stats about the overall performance for each ~~q~~ question
- what preparations were made by the students BEFORE the lectures

professor

ZANDER INTERVIEW

③ Customisation (colors, interface) → light/dark mode
↳ Not much (no specific colors or fonts)
→ Default design that repeats & is familiar when used
↳ pictures
→ ~~Book~~ / List ≡ complete overview
one at a time

④ Anything you like about MOODLE?
↳ the only integrated system in uni
↳ used as little as possible
limits the thinking
WANT: to focus directly on the content

⑤ Accessibility
HIS SIDE: he writes software, doesn't need specific things
→ No mathematical support for him; future integration
(x^2 formatted x^2)

HOW STUDENTS CONDUCT QUIZZES: → reveal the right answers only at the end
!! learn about what they need
increase in the font size, change the contrast
auto translation, contents spoken to them
ability to leave feedback at the end of the quiz (like self evaluation kinda)

⑥ Key values
store the quiz source code together with the lecture material
→ using Markdown to make the quizzes
↳ multiple types of questions (fill in the gaps, multiple choice, attachments, visual support - pictures)
→ in the BG AI telling them feedback (future feature) nice to have